

TECHNICAL RESEARCH ARTICLE

Embedding Model Selection for Persian/English Retrieval-Augmented Generation

A measured comparison of BAAI/bge-m3, intfloat/multilingual-e5-small,
and intfloat/multilingual-e5-base on retrieval quality, cross-lingual
accuracy, score separability, and CPU inference cost

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ABSTRACT

This article reports a controlled, offline, CPU-only benchmark of three open multilingual text-embedding models — BAAI/bge-m3 , intfloat/multilingual-e5-small , and intfloat/multilingual-e5-base — for use as the retrieval component of a Persian/English Retrieval-Augmented Generation (RAG) system. Using a fixed, versioned six-document corpus (English, Persian, and mixed-language documents) and a 23-query evaluation set spanning same-language, cross-lingual, and out-of-corpus queries, we measure top-1 retrieval accuracy, recall@5, score separability between answerable and unanswerable queries, and CPU embedding/query latency. bge-m3 achieved perfect top-1 accuracy and recall@5 (19/19 answerable queries) and was the only model with a positive separability gap, at a CPU cost 4–11× higher than the two e5 models. We discuss the resulting quality/latency trade-off and recommend bge-m3 for production use in this deployment context, with GPU/hosted serving as the recommended mitigation for its CPU throughput cost.

1. Introduction

Retrieval-Augmented Generation (RAG) systems answer questions by first retrieving relevant passages from a private document store and then conditioning a language model on that retrieved context. The quality of the retrieval step is bounded by the embedding model used to represent both documents and queries as vectors: a weak embedding model degrades every downstream answer regardless of how capable the generation model is. This problem is sharper in bilingual deployments — here, Persian and English — where a query and its answer may not share a single surface token, and where a system that gates generation behind a fixed retrieval-score threshold (to avoid answering from irrelevant context) additionally depends on the embedding model producing a clean, consistent separation between "the answer is in the corpus" and "it is not."

This article presents a single, self-contained benchmark run designed to inform that choice for a local-first, CPU-deployable knowledge assistant. It compares three candidate embedding models already used or considered for such deployments, run against identical documents, identical queries, and identical evaluation code, so that any observed difference reflects the models rather than experimental variance.

2. Methodology

2.1 Models under test

Model	Publisher	Embedding dimension	Pooling strategy used
BAAI/bge-m3	BAAI	1024	CLS-token pooling
intfloat/multilingual-e5-small	intfloat	384	Mean pooling, query: / passage: prefixing
intfloat/multilingual-e5-base	intfloat	768	Mean pooling, query: / passage: prefixing

Pooling and prefixing conventions follow each model's published usage recipe **EXTERNAL**; embedding dimensions were read from each model's local configuration file **MEASURED**. All output vectors were L2-normalized prior to scoring, and all similarity scores reported below are cosine similarity.

2.2 Corpus and query set

The evaluation corpus consists of six documents spanning three language conditions: English-only, Persian-only, and mixed-language (a Persian-dominant document containing inline English technical terms, and an English-dominant document containing embedded Persian script). Documents were sourced as PDF and DOCX files to also exercise the system's real text-extraction path, including a documented Persian-script PDF extraction artifact (word/character spacing degradation from ``pdftotext``) that was deliberately left uncorrected so the benchmark measures embedding behavior against the same degraded input a production deployment would actually encounter.

23 queries were authored against this corpus and manually verified against the source text: 10 same-language queries (query and target document share a dominant language), 9 cross-lingual queries (query language differs from the target document's dominant language, with no shared surface tokens), and 4 out-of-corpus queries on topics manually confirmed absent from all six documents, used to measure false-positive retrieval behavior.

2.3 Pipeline and evaluation procedure

Each document was parsed and chunked using the system's own production text-extraction and chunking logic (unmodified), producing 45 chunks in total across the six documents, at a chunk size of 1024 characters with 128 characters of overlap. Each model embedded the identical 45 chunks and 23 queries. Retrieval was computed as cosine similarity between each query vector and every chunk vector, ranked to a top-5 result set per query.

For each answerable query (i.e. queries with a known expected source document), we recorded whether the top-1 result was chunked from the correct document (*top-1 accuracy*) and whether any of the top-5 results came from the correct document (*recall@5*). For out-of-corpus queries, we recorded the top-1 similarity score as a measure of false-positive risk. *Separability gap* is defined as the minimum top-1 score among answerable queries minus the maximum top-1 score among out-of-corpus queries: a positive value means a single fixed similarity threshold exists that accepts every genuine answer while rejecting every out-of-corpus query; a negative value means no such threshold exists, and any fixed cutoff must trade off false accepts against false rejects.

2.4 Benchmark environment and hardware

Parameter	Value
CPU	Intel Core i5-6200U (2 physical cores / 4 threads), CPU-only inference
Inference runtime	ONNX Runtime, <code>CPUExecutionProvider</code> , 4 intra-op threads
Network	Fully offline — no model downloads or network access during the run
Execution order	Sequential: one model loaded, benchmarked, and released before the next started
Chunking	1024-character chunks, 128-character overlap (production defaults)
Batch size	8 texts per embedding batch

Models were embedded through ONNX Runtime rather than a PyTorch/sentence-transformers runtime, reflecting the lighter dependency surface of a bundled, CPU-only deployment. Latency figures below are therefore representative of this runtime and this hardware, not of GPU-served or vendor-hosted inference of the same models.

3. Results

3.1 Retrieval quality

Model	Top-1 accuracy	Recall@5	Same-language acc.	Cross-lingual acc.
bge-m3	100.0% (19/19)	100.0%	100.0%	100.0%
multilingual-e5-small	94.7% (18/19)	94.7%	100.0%	88.9%
multilingual-e5-base	89.5% (17/19)	100.0%	100.0%	77.8%

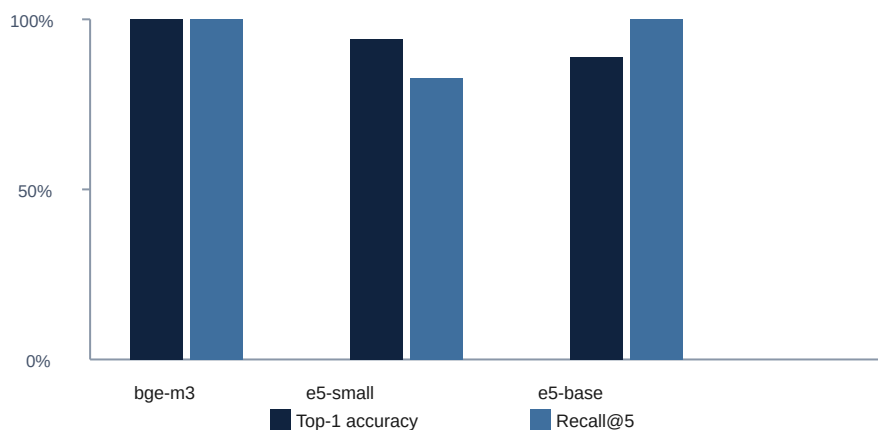


Figure 1. Top-1 retrieval accuracy and recall@5 across the 19 answerable queries, per model.

MEASURED

3.2 Cross-lingual retrieval failures

All three models retrieved correctly on every same-language query. Errors were confined to the cross-lingual category. Both e5 models missed query [q11](#) (a Persian query targeting an English-only document on compound interest); e5-base additionally missed [q17](#) (an English query targeting a Persian-dominant document, which itself contains the English phrase "artificial intelligence" inline — noted in the query set as a partially-shared-surface-term case). bge-m3 made no errors in either query set.

3.3 Score separability

Model	Min top-1 score, answerable	Max top-1 score, out-of-corpus	Separability gap
bge-m3	0.487	0.462	+0.024
multilingual-e5-small	0.779	0.826	-0.047
multilingual-e5-base	0.737	0.786	-0.049

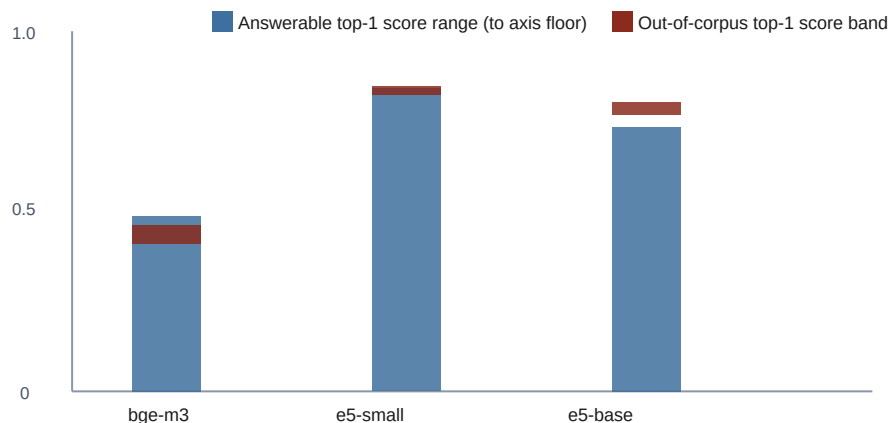


Figure 2. Answerable-query top-1 score range versus out-of-corpus top-1 score band, per model. For bge-m3 the out-of-corpus band (red) sits below the answerable range (blue); for both e5 models the out-of-corpus band overlaps the bottom of the answerable range — the visual signature of a negative separability gap. MEASURED

3.4 CPU performance

Model	Ingestion (ms/chunk)	Ingestion throughput	Query latency (ms)
bge-m3	2604.9	0.38 chunks/s	118.1
multilingual-e5-small	241.4	4.14 chunks/s	12.1
multilingual-e5-base	642.8	1.56 chunks/s	40.2

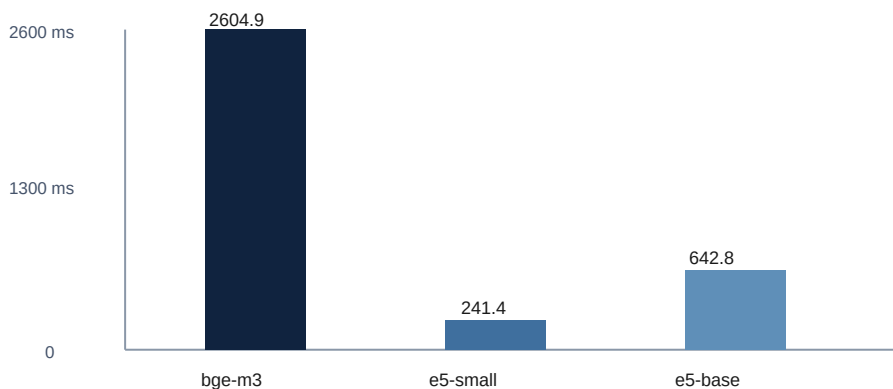


Figure 3. Mean ingestion cost per chunk (milliseconds, wall-clock), ONNX Runtime CPU inference, batch size 8.

MEASURED

4. Quality/Speed Trade-off Analysis

The three models separate cleanly along two axes that move in opposite directions. bge-m3 dominates on every quality metric measured — perfect top-1 accuracy, perfect recall@5, perfect cross-lingual accuracy, and the only positive separability gap — while being 4.1× to 10.8× slower to embed per chunk than e5-base and e5-small respectively, and 2.9× to 9.8× slower per query. e5-small is the fastest model tested by a wide margin but has the weakest cross-lingual accuracy and the largest-magnitude negative separability gap.

The separability gap is the more consequential of the two axes for a system that gates generation behind a fixed similarity threshold to prevent answering from irrelevant context. A positive gap (bge-m3) means one threshold value exists that accepts every genuine answer and rejects every out-of-corpus query in this evaluation set. A negative gap (both e5 models) means the answerable and out-of-corpus score distributions overlap: no fixed threshold can simultaneously avoid rejecting a valid answer and accepting an ungrounded one. On this evidence, adopting either e5 model without re-deriving and accepting a weaker threshold would trade a demonstrated groundedness property for an unverified one.

CPU latency, in contrast, is a deployment-tier cost rather than a correctness property: it can be mitigated by GPU-served or hosted inference of the same model without changing which model is used or re-validating retrieval quality.

5. Production Recommendation

For a Persian/English RAG deployment that depends on a fixed retrieval-score threshold to enforce groundedness, **BAAI/bge-m3** is the recommended embedding model based on this benchmark. It is the only one of the three models tested with a positive separability gap and the only one with perfect cross-lingual retrieval accuracy on this evaluation set. Its CPU embedding cost is substantially higher than the alternatives, but this is addressable through GPU-served or hosted inference of the same model without sacrificing any of the measured quality result — whereas substituting a faster model would require accepting a measurably weaker groundedness guarantee and materially worse cross-lingual retrieval. Where an application does not require a fixed-threshold groundedness gate and can tolerate a small, measured reduction in cross-lingual accuracy, multilingual-e5-small offers markedly lower CPU latency and remains a defensible choice.

6. Limitations and Threats to Validity

- **Corpus size.** Six documents and 23 queries (19 answerable, 4 out-of-corpus) is a small evaluation set. Reported accuracies and separability gaps may shift, in either direction, at larger scale.
- **Topic coverage.** Each topic in the corpus is represented by exactly one document, so the benchmark does not test ranking among multiple same-topic documents of varying relevance.
- **Language-mixing patterns.** Only two mixed-language patterns are represented (Persian-dominant with inline English terms, and English-dominant with embedded Persian script). Real-world documents may mix languages differently, e.g. at the sentence level.
- **Document complexity.** No scanned/image-based PDFs, tables, or complex page layouts are represented in the corpus.

- **Known extraction artifact.** Text extracted from the two Persian-script PDF documents shows a previously documented spacing degradation from the PDF-to-text tool used; this was left uncorrected so results reflect production extraction behavior, but it may understate the achievable retrieval quality on cleanly extracted Persian PDF text.
- **Hardware and runtime specificity.** CPU latency figures are specific to the tested processor and to ONNX Runtime CPU inference; they do not generalize to GPU or vendor-hosted serving of the same models.
- **Single benchmark run.** Each model was embedded once per corpus/query set, without repeated trials to characterize latency variance.

7. References

1. BAAI, *bge-m3: Multi-Lingual, Multi-Functionality, Multi-Granularity Text Embeddings*, model card and documentation.

EXTERNAL, MODEL PROVENANCE ONLY

2. Wang et al. (intfloat), *Multilingual E5 Text Embeddings*, model cards for `multilingual-e5-small` and `multilingual-e5-base`.

EXTERNAL, MODEL PROVENANCE ONLY

3. ONNX Runtime, *CPU Execution Provider documentation*.

EXTERNAL, RUNTIME PROVENANCE ONLY

All quantitative results reported in Sections 3–4 of this article are original measurements produced by the benchmark described in Section 2 and are labeled **MEASURED** throughout; statements drawn from model documentation rather than this benchmark are labeled **EXTERNAL**.

Independent technical article. All measurements were produced by a single, offline, CPU-only benchmark run described in Section 2; no external benchmark results, published leaderboard scores, or unverified claims about model behavior are presented as measured data in this article.